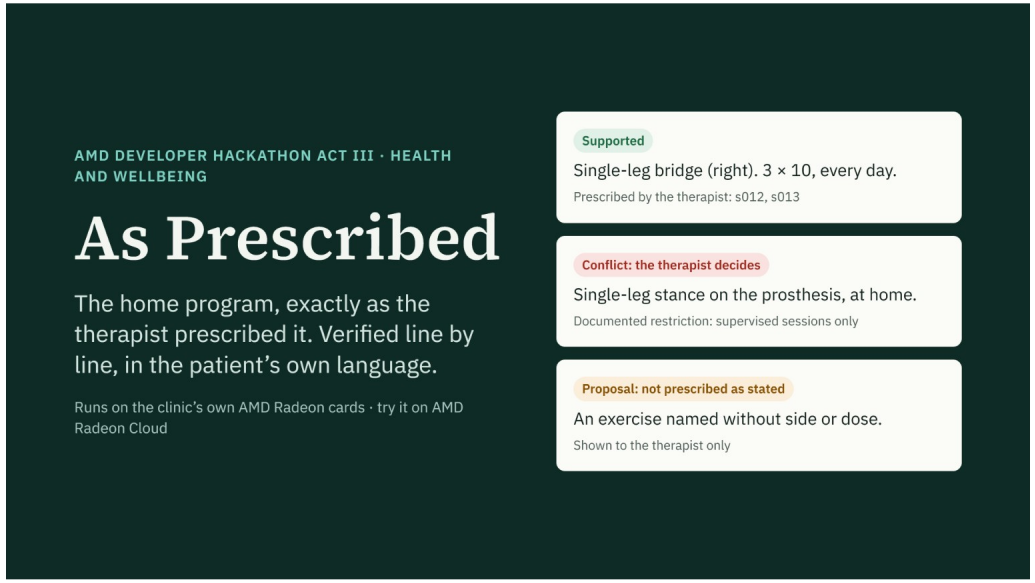


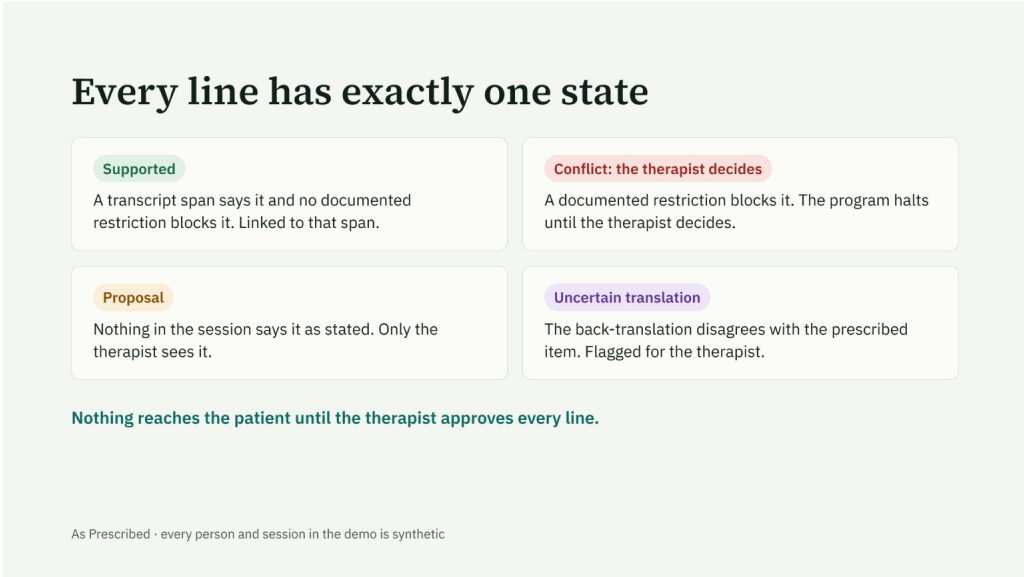
As Prescribed turns a physiotherapy visit into the patient's home program in their own language, and verifies every line against what the therapist actually said. It runs on the clinic's own AMD Radeon cards; judges can try the same pipeline live on AMD Radeon Cloud. All data in the demo is synthetic.



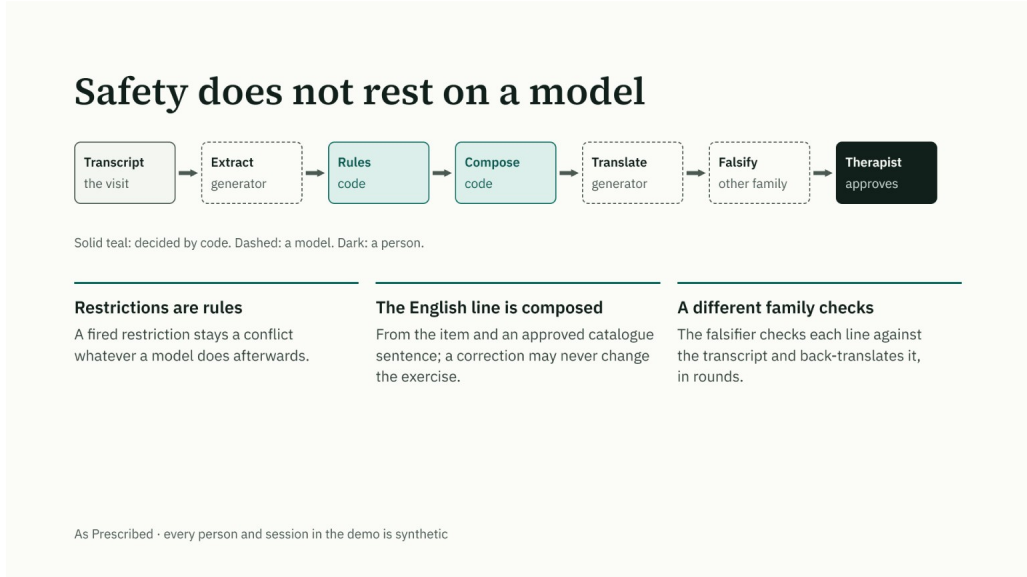
Picture a patient who does not speak the clinic’s language. They leave with a home program they cannot read, or with a machine translation nobody in the room can check. A wrong side or a dropped caution is a real safety problem. And a free-writing AI makes it worse: it adds plausible exercises the therapist never prescribed.



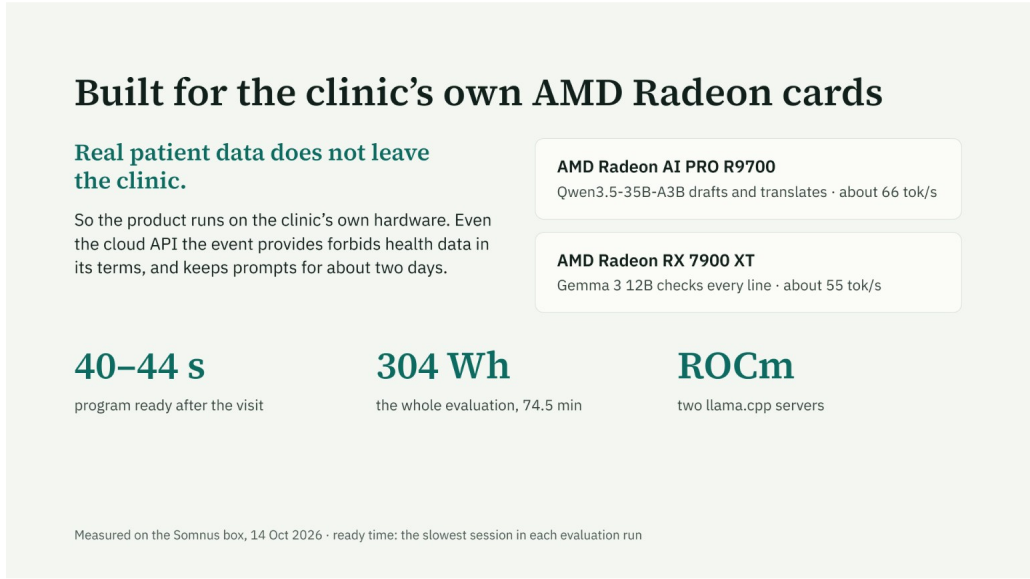
This is the boundary, and it is the first safety measure. The system checks three things: fidelity to what was said, the patient’s documented restrictions, and the translation. It never decides whether an exercise suits the injury.



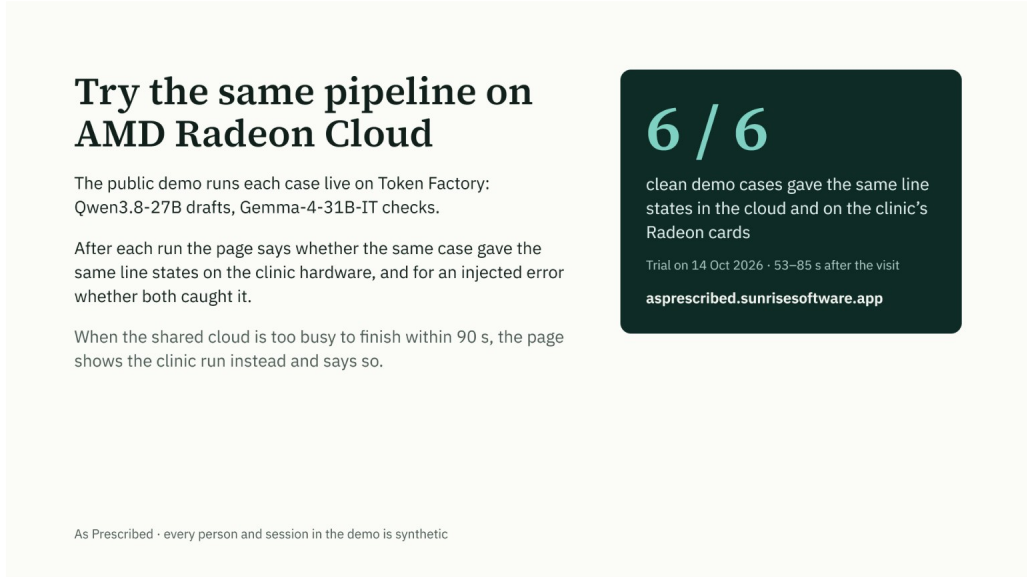
During the visit the transcript streams in and the program is drafted as the therapist speaks. When the visit ends, every line is in one of four states. A conflict halts the program; a proposal is never released without the therapist’s explicit approval.



A generator model extracts what was prescribed and translates. Everything that can be decided without a model is decided by code: the rule engine checks documented restrictions, and the English line is composed from the item, never written by a model. A falsifier from a different model family checks every line against the transcript and back-translates it, in rounds. We learned the hard way that a model checking its own family passes its own family’s errors.



This is the deployment that matters. Real patient data cannot go to a cloud API; even the event’s own API terms forbid health data. So As Prescribed runs on two Radeon cards in the clinic: a Radeon AI PRO R9700 drafts and translates with Qwen3.5, and a Radeon RX 7900 XT checks every line with Gemma 3. The program is ready about 40 seconds after the visit, and our whole evaluation used 304 watt-hours.



Judges can try it without our hardware. The public demo runs the same pipeline live on AMD Radeon Cloud through Token Factory, and after each run it shows how the same case went on the clinic hardware. In our trial all six clean cases gave the same line states in both places. The shared cloud is sometimes busy; then the page shows the clinic run and says so, rather than keep anyone waiting.

Measured, and reported as it fell

32 injected errors of six types and 9 clean runs, in a plan sealed by its sha256 before the first run. Detection is read before any correction.

Measure	Run 1, pre-registered	Run 2, after fixes	Threshold
Recall, all 32 errors	0.59, below	0.72, below	0.80
Recall, the 18 only a model can catch	0.28, below	0.50, below	0.60
Recall, the 14 the rules and checks catch	1.00	1.00	none
False alarms on clean lines	16.2 %, above	6.9 %, met	15 %
Ready after the visit, slowest session	40 s, met	44 s, met	90 s

The recall gate failed in both runs. Every documented restriction was caught.

Local mode on the Radeon cards · run 2 was made after reading run 1 and is not an independent test · raw runs in eval/

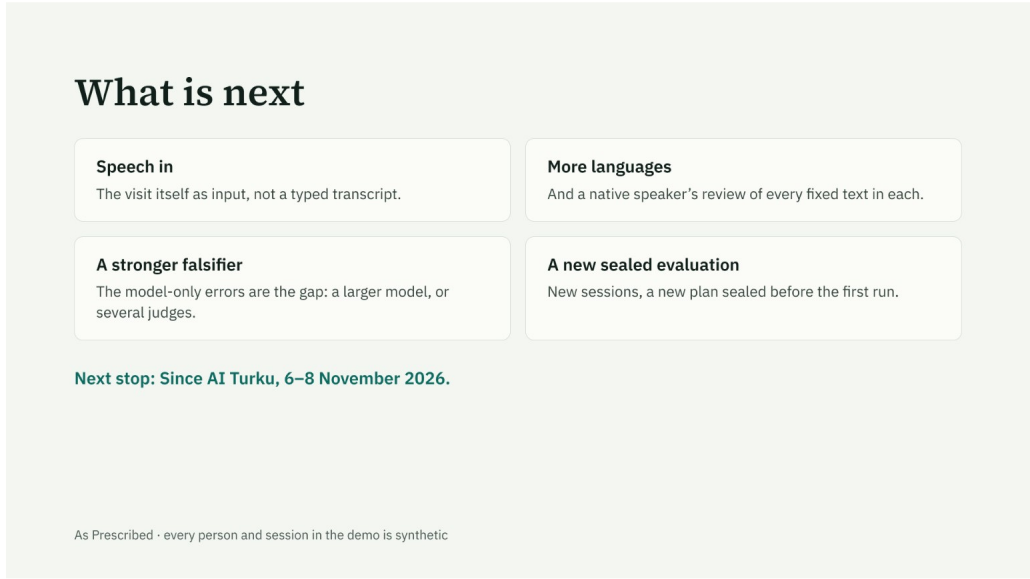
We sealed the evaluation before the first run and report it as it fell. The deterministic layer caught every error it is meant to catch, including every contraindicated exercise. The model falsifier found about a quarter of the errors only a model can find in the pre-registered run, and half after fixes. That second run was made after reading the first, so it is not an independent test. The recall gate failed in both runs, and we say so.

The track’s five safety questions

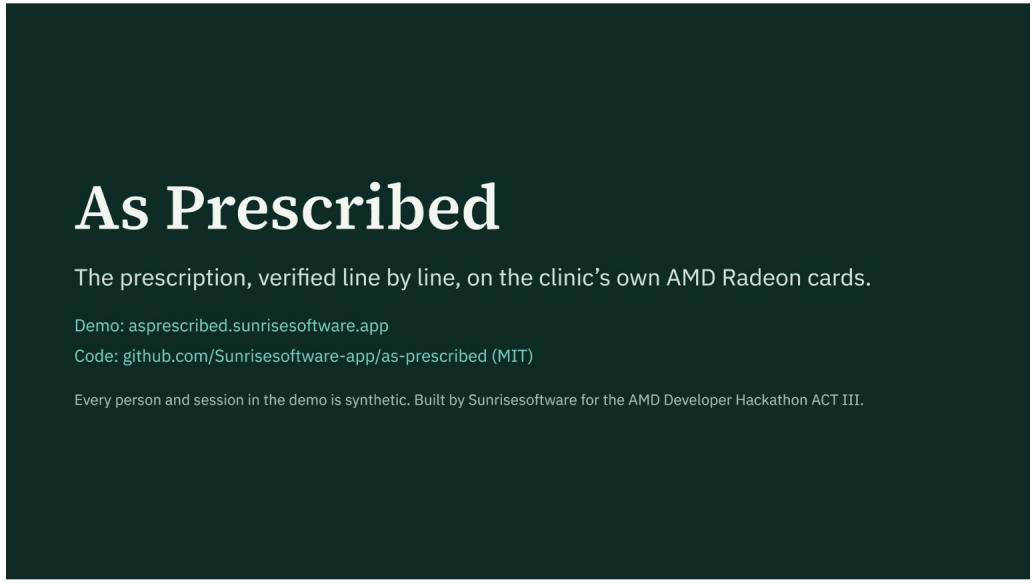
Question	As Prescribed
Incorrect or unsupported responses	A line no transcript span supports is a proposal, shown to the therapist only.
Sensitive information	Synthetic data only, no free-text input in the demo; in a clinic, everything stays on its own hardware.
Uncertain results	Uncertain translations and the falsifier’s remaining objections are shown, never hidden.
High-risk decisions	A documented restriction halts the program; the therapist decides.
Human review	Nothing reaches the patient without the therapist’s approval.

As Prescribed · every person and session in the demo is synthetic

The track asks five safety questions. Our answers all come back to the boundary and to the therapist: unsupported lines become proposals, uncertainty is shown, a documented restriction halts the program, and nothing reaches the patient without the therapist’s approval.



Next: speech as the input, more languages with a native speaker's review of every fixed text, a stronger falsifier for the errors only a model can catch, and a new sealed evaluation on new sessions. The same code base continues at Since AI Turku in November.



Try it at asprescribed.sunrisesoftware.app; the code is open under MIT. Thank you.